PHY 208 Equation Sheet

Constants

$$k = Coulomb\ Constant = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9\, Nm^2C^{-2}$$

$$\epsilon_0 = vacuum\ permittivity = 8.85 \times 10^{-12}\, N^{-1}C^2m^{-2}$$

$$e = 1.60 \times 10^{-19}\,\mathrm{C}\,(\text{charge of electron})$$

Force and Electric Field

$$F = k\frac{|q_1 q_2|}{r^2}\,(Coulomb's\ law)$$

$$\vec{E} = \frac{\vec{F}}{q_0}\,(Electric\ field)$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r}\,(E\ field\ from\ a\ point\ charge)$$

$$\lambda \equiv \frac{Q}{L}\,(Linear\ charge\ density)$$

$$E = \frac{1}{2\pi\epsilon_0}\frac{\lambda}{r}\,(Efield\ from\ an\ infinite\ wire)$$

$$\sigma \equiv \frac{Q}{A}\,(Surface\ charge\ density)$$

$$E = \frac{\sigma}{2\epsilon_0}\,(E\ field\ from\ infinite\ sheet\ of\ charge)$$

$$\rho \equiv \frac{Q}{V}(Volume\ charge\ density)$$

$$|\tau| = pE\sin\phi = qdE\sin\phi = (Torque\ on\ an\ electric\ dipole)$$

$$U = -\vec{p}\cdot\vec{E}(Potential\ Energy\ for\ an\ electric\ dipole)$$

$$\Phi_E = \int \vec{E}\cdot d\vec{A} = \int E\cos\phi\ dA\,(Electric\ flux)$$

$$\Phi_E = \oint \vec{E}\cdot d\vec{A} = \frac{Q_{encl}}{\epsilon_0}\,(Gauss's\ law)$$

$$E_\perp = \frac{\sigma}{\epsilon_0}(E\ field\ at\ surface\ of\ conductor)$$

Electric Potential and Potential Energy

$$W_{a\to b} = U_a - U_b = -(U_b - U_a) = -\Delta U$$

$$(Work\ by\ conservative\ force)$$

$$\Delta U = q\,\Delta V\,(E\ potential\ energy)$$

$$\Delta V = -\int \vec{E}\cdot d\vec{l} = -\int E\cos\phi dl$$

$$V = E\,d\,(E\ potential\ in\ uniform\ E)$$

$$V = \frac{1}{4\pi\epsilon_0}\frac{q}{r}\,(E\ potential\ due\ to\ a\ point\ charge)$$

$$V = \frac{1}{4\pi\epsilon_0}\Sigma_i\frac{q_i}{r_i}\,(E\ potential\ of\ multiple\ charges)$$

$$V = \frac{1}{4\pi\epsilon_0}\int\frac{dq}{r}$$

$$(E\ potential\ due\ to\ continues\ charge\ dist.)$$

$$U = \frac{1}{4\pi\epsilon_0}\frac{qq_0}{r}$$

$$(Electric\ potential\ energy\ of\ two\ charges)$$

$$E_x = -\frac{\partial V}{\partial x},\ E_y = -\frac{\partial V}{\partial y},\ E_z = -\frac{\partial V}{\partial z}$$

Capacitance

$$C = \frac{Q}{\Delta V}\,(Capacitance)$$

$$C = \frac{\epsilon_0 A}{d}\,(Capacitance\ of\ parallel\ plates)$$

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + ...\,(capacitors\ in\ series)$$

$$C_{\text{eq}} = C_1 + C_2 + ...\,(capacitors\ in\ parallel)$$

$$U = \frac{Q^2}{2C} = \frac{1}{2}CV^2\,(energy\ in\ capacitor)$$

$$C_d = KC\,(capacitor\ filled\ w.\ dielectric)$$

Current and Resistors

$$I = \frac{dQ}{dt} = n|q|v_dA$$

$$\vec{J} = nq\vec{v_d}\,(vector\ current\ density)$$

$$V = I\,R\,(Ohm's\ law)$$

$$R = \frac{\rho L}{A}\,(resistance) \quad \rho = \frac{E}{J}\,(resistivity)$$

$$P = I^2R = I\,V = \frac{V^2}{R}\,(power\ delivered\ to\ resistor)$$

$$R_{\text{eq}} = R_1 + R_2 + ...\,(resistors\ in\ series)$$

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + ...\,(resistors\ in\ parallel)$$

$$\sum I = 0\,(Kirchhoff's\ current\ junction\ rule)$$

$$\sum V = 0\,(Kirchhoff's\ voltage\ loop\ rule)$$

$$I(t) = \frac{V}{R}\,e^{-t/\tau},\ \tau = RC\,(current\ of\ DC\ circuit)$$

$$Q(t) = Q(t=\infty)(1 - e^{-t/\tau})\,(charging\ a\ capacitor)$$

$$Q(t) = Q(t=0)e^{-t/\tau}\,(discharging\ a\ capacitor)$$

Constants

$c = 3 \times 10^8 m/s$ (speed of light in vacuum)

$\mu_0 = vacuum\ permeability = 4\pi \times 10^{-7}\,\mathrm{T\,m\,A^{-1}}$

Magnetic forces and torque

$\vec{F} = q\,\vec{v} \times \vec{B}\,(magnetic\ force)$

$\vec{F} = q\,(\vec{v} \times \vec{B} + \vec{E})\,(magnetic\ and\ electric\ force)$

$\vec{F} = I\vec{l} \times \vec{B}\,(magnetic\ force\ on\ conductor)$

$\vec{\mu} = I\,\vec{A}\,(magnetic\ moment)$

$\vec{\tau} = \vec{\mu} \times \vec{B}\,(magnetic\ torque)$

$\tau = IBA\sin\phi\,(magnitude\ of\ magnetic\ torque)$

$U = -\vec{\mu} \cdot \vec{B} = -\mu B\cos\phi\,(magnetic\ potential\ energy)$

$F = |q|vB = m\frac{v^2}{R}\ and\ \omega = \frac{v}{R}$

$R = \frac{mv}{|q|B}\,(radius\ of\ circular\ orbit\ in\ magnetic\ field)$

Induction

$\varepsilon_2 = -M\frac{di_1}{dt}\ and\ \epsilon_1 = -M\frac{di_2}{dt}(Mutually\ induced\ emf)$

$M = \frac{N_2\Phi_{B_2}}{i_1} = \frac{N_1\Phi_{B_1}}{i_2}\,(Mutual\ inductance)$

$L = \frac{N\Phi_B}{i}\,(Self-inductance)$

$\epsilon = -L\frac{di}{dt}\,(Self-induced\ emf)$

$U = \frac{1}{2}LI^2\,(Energy\ stored\ in\ an\ inductor)$

$u = \frac{B^2}{2\mu_0}\,(Magnetic\ Energy\ density\ in\ vaccum)$

$i = \frac{\varepsilon}{R}(1 - e^{-\frac{R}{L}t})\,(current\ in\ RL\ circuit)$

$i = I_0(e^{-\frac{R}{L}t})\,(current\ decay\ in\ RL\ circuit)$

$\tau = \frac{L}{R}\,(time\ constant\ in\ RL\ circuit)$

$q = Q\cos(\omega t + \phi)\,(charge\ in\ capacitor\ in\ LC\ circuits)$

$i = -\omega Q\sin(\omega t + \phi)\,(current\ in\ capacitor\ in\ LC\ cir.)$

$\omega = \sqrt{\frac{1}{LC}}\,(Angular\ frequency\ in\ LC\ circuits)$

$q = Ae^{-\frac{R}{2L}t}\cos(\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}t + \phi)\,(cap.\ charge\ in\ LRC)$

$\omega' = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}\,(Ang.\ freq.in\ LRC\ cir.)$

Sources of Magnetic Field

$\vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v} \times \hat{r}}{r^2}\,(B\ from\ moving\ charge)$

$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,d\vec{l} \times \hat{r}}{r^2}\,(B\ from\ current)$

$B = \frac{\mu_0 I}{2\pi r}\,(B\ from\ a\ long\ wire)$

$\frac{F}{L} = \frac{\mu_0 II'}{2\pi r}\,(Magnetic\ force\ per\ unit\ length)$

$B_x = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}}\,(Magnetic\ field\ on\ axis\ of\ loop)$

$B_x = \frac{\mu_0 NI}{2a}\,(Magnetic\ field\ at\ center\ of\ N\ loop)$

$B = \mu_0 nI\ (B\ by\ loops\ w.\ n\ turns\ per\ unit\ length)$

$B = \frac{\mu_0 NI}{2\pi r}\,(B\ of\ toroidal\ solenoid)$

$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\mathrm{encl}}\ (Ampere's\ law)$

$\mu = K_m\mu_0\,(permeability\ of\ materials)$

$\chi_m = K_m - 1\,(magnetic\ susceptibility)$

$\Phi = BA\cos\phi\,(magnetic\ flux)$

$\varepsilon = -N\frac{d\Phi_B}{dt}\,(Faraday's\ law)$

$\oint \vec{E} \cdot d\vec{l} = -N\frac{d\Phi_B}{dt}\,(Faraday's\ law\ for\ int.\ path)$

$\varepsilon = vBL\,(Motional\ emf)$

$\varepsilon = \oint (\vec{v} \times \vec{B}) \cdot d\vec{l}\,(Motional\ emf,\ general\ case)$

$i_D = \epsilon_0\frac{d\Phi_E}{dt}\,(Displacement\ current)$

$\oint \vec{B} \cdot d\vec{A} = 0\ (Gauss's\ law\ for\ B)$

$\oint \vec{B} \cdot d\vec{l} = \mu_0(i_C + \epsilon_0\frac{d\Phi_E}{dt})\ (Ampere'slaw)$

Kinematics

$$x = x_0 + v_0 t + \frac{1}{2}at^2$$

$$a_{rad} = \frac{v^2}{r}$$

$$F = ma$$

$$F = ma_{rad} = m\frac{v^2}{R} (Uniform\ circular\ motion)$$

$$K = \frac{1}{2}mv^2\,(Kinetic\ Energy)$$

$$U = mgh\,(Gravitational\ Potential\ Energy)$$

EM waves

$$c = f\lambda = \frac{\omega}{k}\,(EM\ wave\ speed)$$

$$\omega = 2\pi f\,(angular\ frequency\ v.s\ frequency)$$

$$k = \frac{2\pi}{\lambda}\,(wave\ number\ v.s\ wavelength)$$

$$E = cB\,(EM\ wave\ in\ vacuum)$$

$$B = \epsilon_0\mu_0 cE\,(EM\ wave\ in\ vacuum)$$

$$c = \frac{1}{\sqrt{\epsilon_0\mu_0}}\,(speed\ of\ EM\ wave\ in\ vacuum)$$

$$\vec{E}(x,t) = \hat{j}E_{max}\cos(kx - \omega t)\,(sinusoidal\ E\ field)$$

$$\vec{B}(x,t) = \hat{k}B_{max}\cos(kx - \omega t)\,(sinusoidal\ B\ field)$$

$$u = \epsilon_0 E^2\,(EM\ wave\ energy\ density)$$

$$\vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B}\,(Poynting\ vector\ in\ vacuum)$$

$$I = S_{av} = \frac{E_{max}B_{max}}{2\mu_0}\,(Intensity\ of\ EM\ wave\ in\ vacuum)$$

$$I = S_{av} = \frac{1}{2}\epsilon_0 cE_{max}^2\,(Intensity\ of\ EM\ wave\ in\ vacuum)$$

$$P = \oint \vec{S}\cdot d\vec{A}\,(energy\ flow\ per\ unit\ time)$$

$$\frac{1}{A}\frac{dp}{dt} = \frac{S}{c} = \frac{EB}{\mu_0 c}(Flow\ rate\ of\ EM\ momentum)$$

$$P_{rad} = \frac{S_{av}}{c} = \frac{I}{c}(Rad.\ pressure,\ wave\ tot.\ absorbed)$$

$$P_{rad} = \frac{2S_{av}}{c} = \frac{2I}{c}(Rad.\ pressure,\ wave\ tot.\ reflected)$$

Optics

$$\lambda = \frac{\lambda_0}{n}$$

$$v = \frac{c}{n}\,(light\ speed\ in\ medium)$$

$$\theta_r = \theta_a\,(law\ of\ reflection)$$

$$n_a\sin\theta_a = n_b\sin\theta_b\,(law\ of\ refraction)$$

$$\sin\theta_{crit} = \frac{n_b}{n_a}\,(total\ internal\ reflection)$$

$$n = \frac{R}{A}\,(Real\ to\ apparent\ ratio)$$

$$I = I_{max}\cos^2\phi\,(Malus's\ law)$$

$$\tan\theta_p = \frac{n_b}{n_a}\,(Brewsters's\ law\ for\ polarization)$$

$$m = \frac{y'}{y} = -\frac{s'}{s}\,(Lateral\ Magnification)$$

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}\,(Object\ -image\ relationship)$$

$$f = \frac{R}{2}\,(focal\ length\ of\ spherical\ mirror)$$

$$\frac{n_a}{s} + \frac{n_b}{s'} = \frac{n_b - n_a}{R}\,(Obj/Im\ Rel.\ Spherical\ ref.)$$

$$\frac{n_a}{s} + \frac{n_b}{s'} = 0\,(Apparent\ depth\ v.s\ virtual\ depth)$$

$$\frac{1}{f} = (n-1)(\frac{1}{R_1} - \frac{1}{R_2})\,(Lensmaker's\ equation)$$

$$f - number\ of\ a\ lens = \frac{f}{D}$$

$$M = \frac{\theta'}{\theta} = \frac{y/f}{y/25cm} = \frac{25cm}{f}\,(Angular\ mag.)$$

$$M = m_1 M_2 = \frac{(25cm)s_1'}{f_1 f_2}\,(Microscope\ ang.\ mag.)$$

$$M = \frac{\theta'}{\theta} = \frac{-y'/f_2}{y'/f_1} = -\frac{f_1}{f_2}\,(Telescope\ ang.\ mag.)$$

Interference

$$d\sin\theta = m\lambda,\ m = 0, \pm1, \pm2, ...$$
$$(constructive\ interference)$$

$$d\sin\theta = (m + \frac{1}{2})\lambda,\ m = 0, \pm1, \pm2, ...$$
$$(destructive\ interference)$$

$$y_m = R\frac{m\lambda}{d},\ m = 0, \pm1, \pm2, ... \quad (bright\ fringes)$$

$$I = I_0 \cos^2\left(\frac{\phi}{2}\right),\ \phi = \frac{2\pi d\sin\theta}{\lambda} \quad (intensity)$$

$$\phi = 0 \quad (phase\ shift\ when\ wave\ reflected\ by\ interface\ with\ lower\ index\ of\ refraction)$$

$$\phi = \pi \quad (phase\ shift\ when\ wave\ reflected\ by\ interface\ with\ higher\ index\ of\ refraction)$$

$$2t = m\lambda,\ m = 0, 1, 2, ...(constructive\ reflection\ from\ thin\ film\ with\ no\ relative\ phase\ shift\ or\ destructive\ reflection\ with\ half\ cycle\ phase\ shift\ \pi)$$

$$2t = (m + \frac{1}{2})\lambda,\ m = 0, 1, 2, ...(destructive\ reflection\ from\ thin\ film\ with\ no\ relative\ phase\ shift\ or\ constructive\ reflection\ with\ half\ cycle\ phase\ shift\ \pi)$$

Diffraction

$$\sin\theta = \frac{m\lambda}{a},\ m = \pm1,\ \pm2,\ \pm3, ...$$
$$(single\ slit\ diffraction)$$

$$I = I_0\left(\frac{\sin\beta}{\beta}\right)^2,\ \beta = \frac{\pi a\sin\theta}{\lambda}(Intensity)$$

$$\theta_{\min} = 1.22\frac{\lambda}{D}$$
$$(angular\ radius\ of\ the\ first\ dark\ ring\ of\ a\ round\ aperture)$$

NuclearPhysics

$$r = r_0 A^{1/3} \quad \text{(nuclear radius)}$$
$$r_0 = 1.2 \times 10^{-15}\ \text{m} = 1.2\ \text{fm}$$
$$V = \frac{4}{3}\pi r_0^3 A \quad \text{(nuclear volume)}$$
$$A = \text{atomic mass}$$
$$E = mc^2 \quad \text{(relativistic energy)}$$
$$E_B = (Nm_n + Zm_H - m_{\text{atom}})c^2$$
(binding energy of a nucleus)
$$1\,u = 931.5\,\text{MeV}/c^2 \quad \text{(atomic mass unit)}$$
$$\text{Binding energy per nucleon} = \frac{E_B}{A}$$
$$\text{Radioactive decay :} \quad \frac{dN}{dt} = -\lambda N$$
$$N(t) = N_0 e^{-\lambda t}$$
$$t_{1/2} = \frac{\ln 2}{\lambda} \quad (Half - life)$$
Energy released in reaction :
$$(\Delta m)c^2 = \left(\sum m_{\text{original}} - \sum m_{\text{products}}\right)c^2$$
$$1\,\text{Curie (Ci)} = 3.7 \times 10^{10} \quad \text{decays/second}$$
Absorbed dose :
$$1\,\text{rad} = 0.01\,\text{J/kg} = 0.01\,\text{Gy}$$
Equivalent dose (Sv)
$$= \text{RBE} \times \text{absorbed dose (Gy)}$$
Equivalent dose (rem)
$$= \text{RBE} \times \text{absorbed dose (rad)}$$